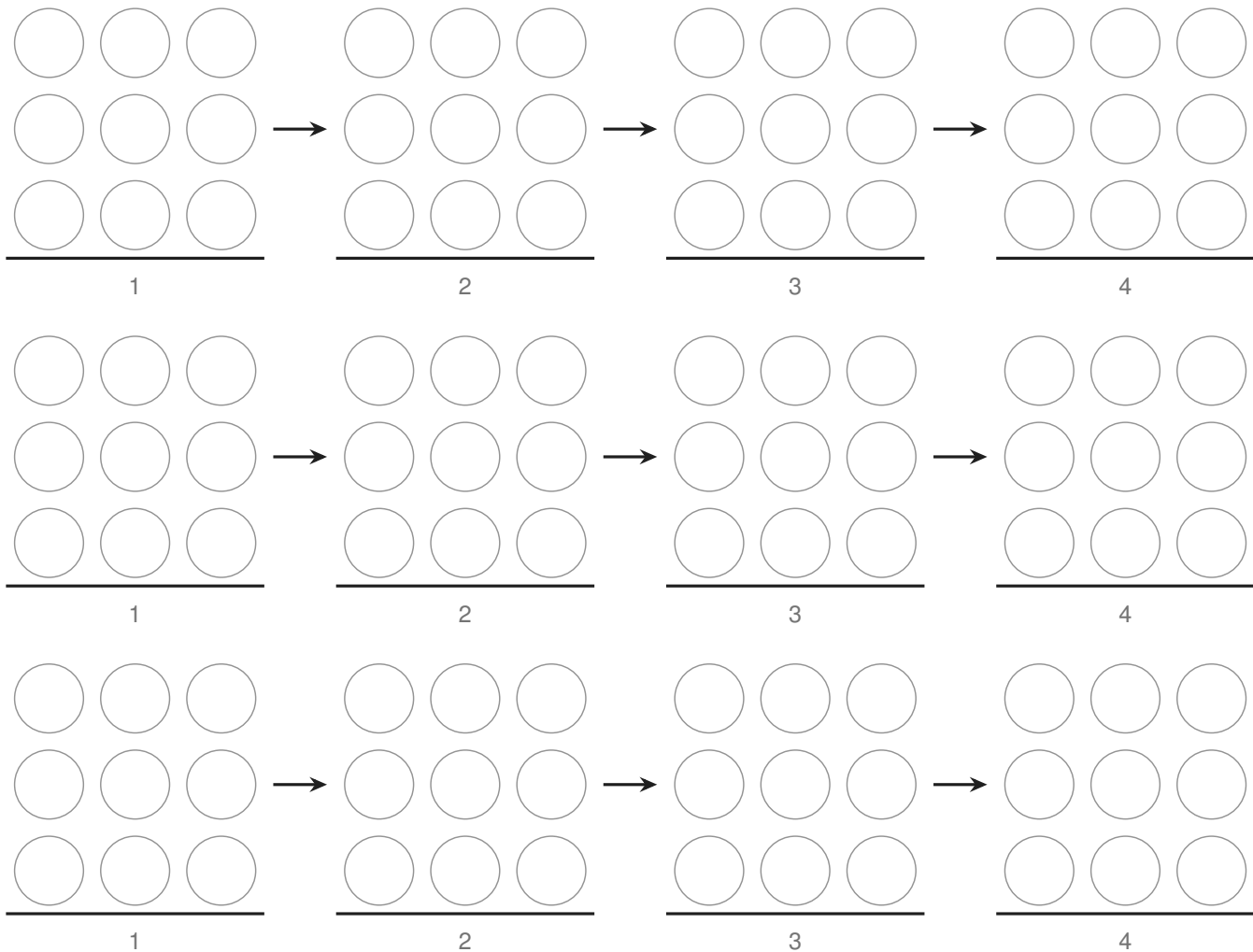
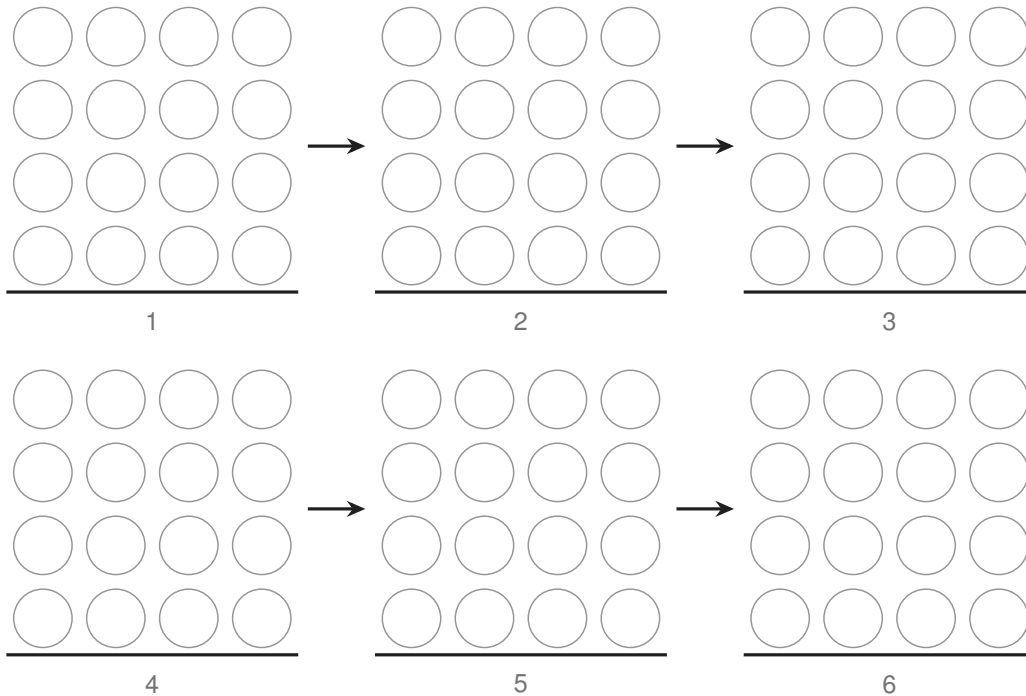


Problem 1: Take 3 counters. Make them into piles any way you like. Put the piles in a row, biggest pile first. In picture 1, color one tower of circles for each pile, as tall as the pile, starting at the bottom line. Now do the move: take one counter from every pile, and put the counters you took together as one new pile. Put the piles in a row again, biggest first, and color them in the next picture. Keep doing the move and coloring the next picture until a picture looks exactly like the one before it. There are three ways to make piles with 3 counters. Use one row of pictures for each way.

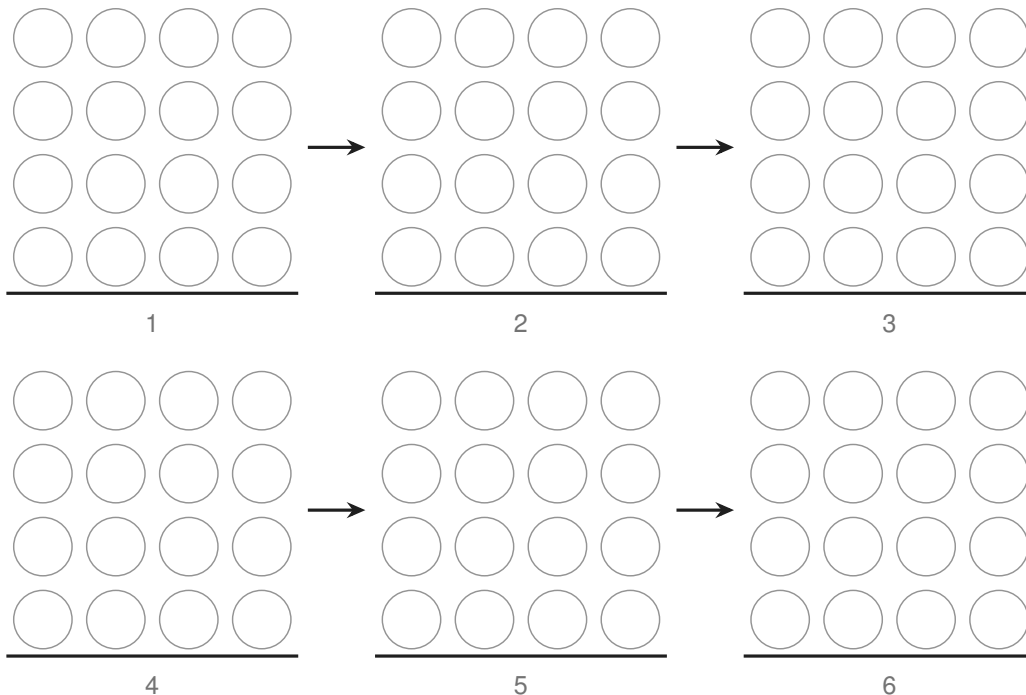


Problem 2: Circle the last picture in each row of Problem 1. Build that shape with your counters. Do the move three times. Say what happens each time.

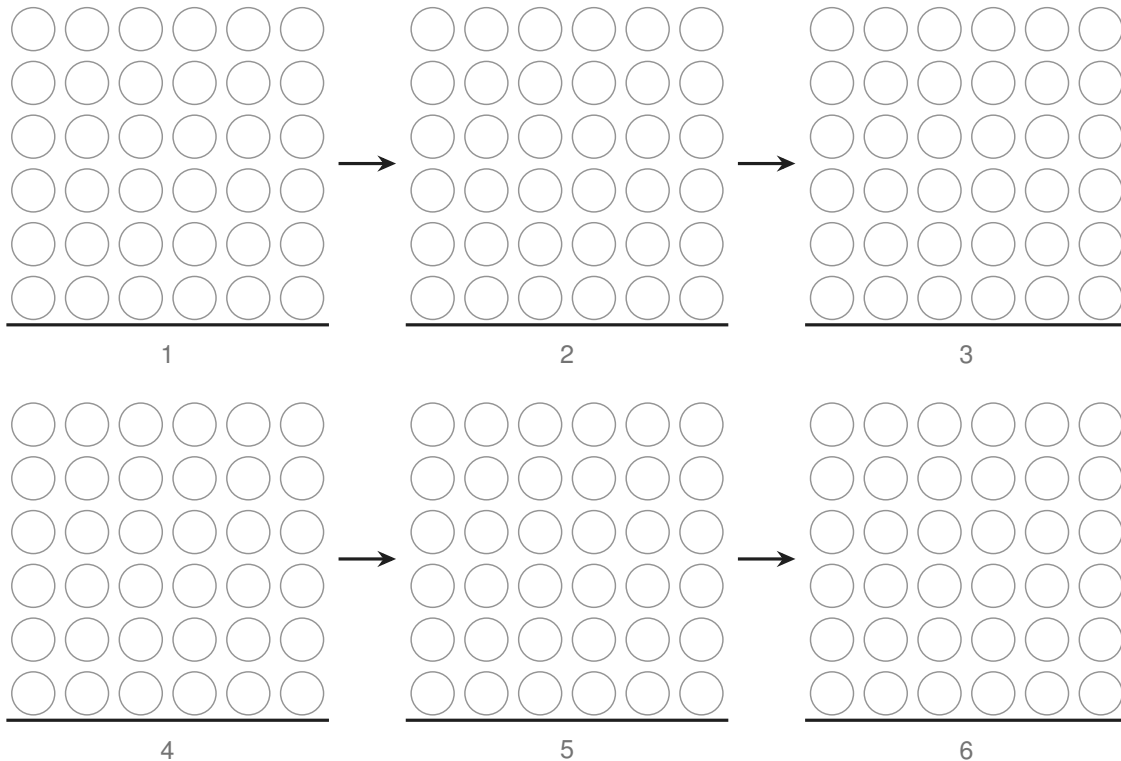
Problem 3: Take 4 counters. Put all 4 in one pile and color it in picture 1: one tower of 4. Do the move and color the piles in picture 2, biggest first. Keep going until all six pictures are colored. Then find two pictures that look the same and circle both.



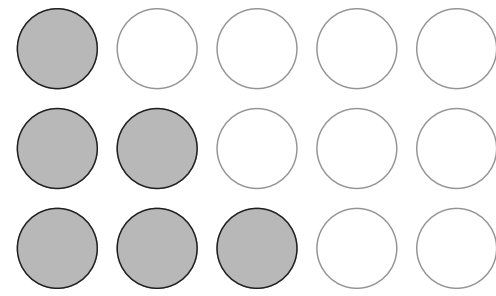
Problem 4: Take 4 counters. Start with four piles of 1 and color them in picture 1. Do the move and color, until all six pictures are colored. Circle two pictures that look the same. Then put a check mark under every picture that also appears somewhere in Problem 3.



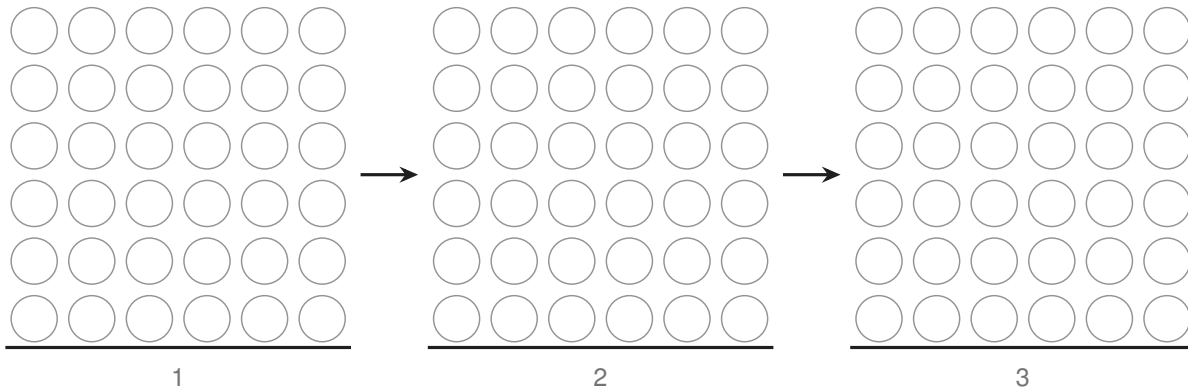
Problem 5: Take 6 counters. Start with all 6 in one pile and color it in picture 1. Do the move, color the piles in picture 2, and keep going. Stop when a picture looks exactly like the one before it. Circle those two pictures. Count the arrows from picture 1 to the first circled picture and write that number in the box.



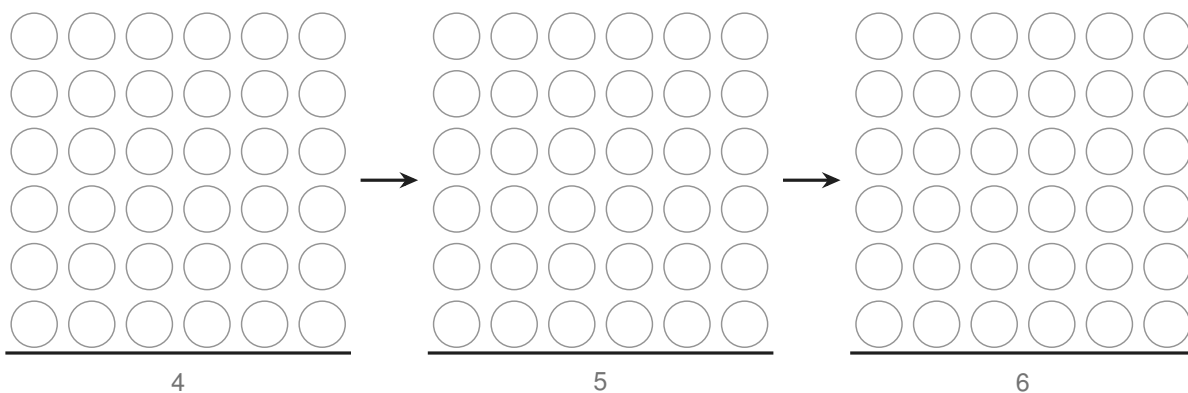
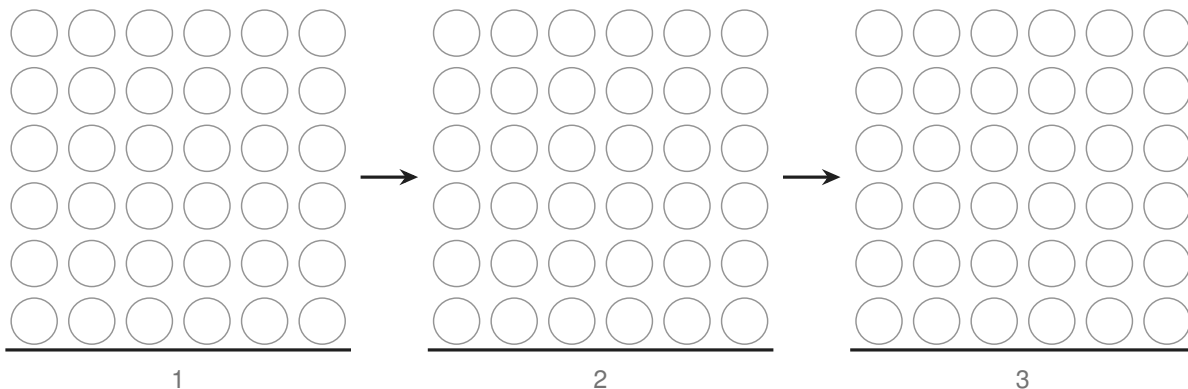
Problem 6: This picture shows the piles 3, 2, 1. Build them with your counters. Now do the move slowly: take the top counter off each pile and put those three counters together as the new pile. In the picture, cross out the top circle of each tower, then color the new pile as a tower in the empty space on the right. Put your real piles in a row, biggest first, and compare them with the towers that are left. Tell your table what you see.



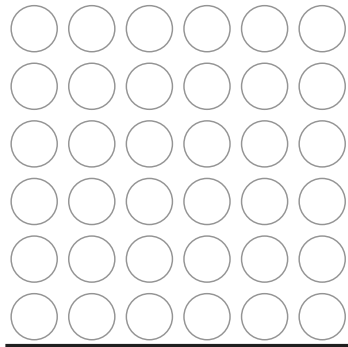
Problem 7: Take 6 counters. Start with six piles of 1 and color them in picture 1. Do the move and color picture 2. Do the move again and color picture 3. Now look back at Problem 5. Find the picture there that looks exactly like your picture 3 and point to it. Say what will happen next, then check with your counters.



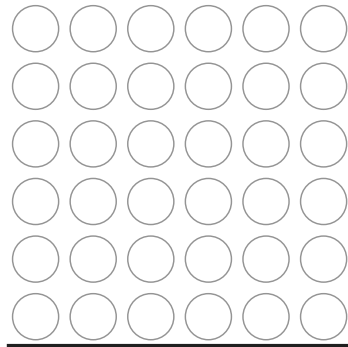
Problem 8: Take 6 counters. Start with two piles of 3 and color them in picture 1. Do the move and color, until a picture looks exactly like the one before it. Circle those two pictures. Count the arrows from picture 1 to the first circled picture and write that number in the box.



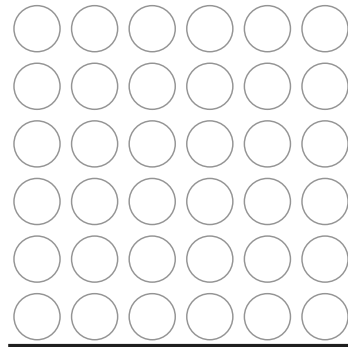
Problem 9: Take 6 counters. Make piles in a way that nobody at your table has used yet. Color them in picture 1. Do the move and color, until a picture looks exactly like the one before it. Circle those two pictures. Count the arrows from picture 1 to the first circled picture and write that number in the box.



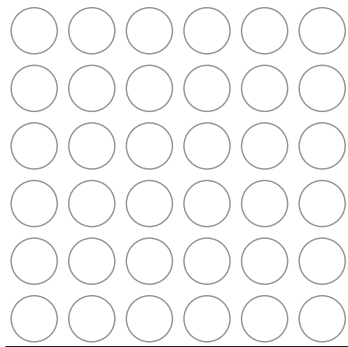
1



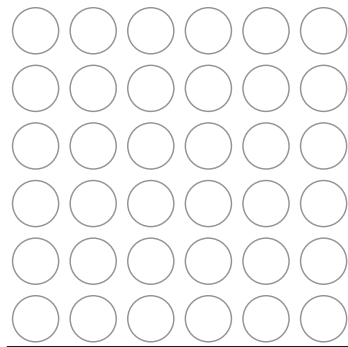
2



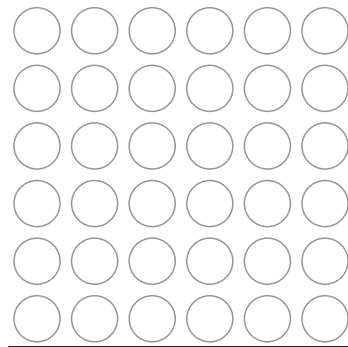
3



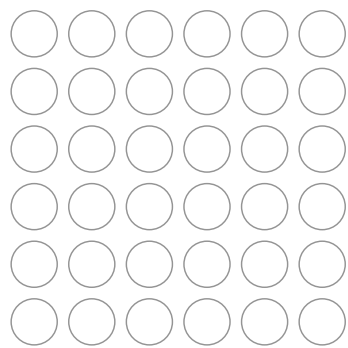
4



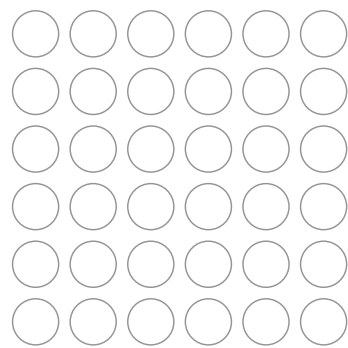
5



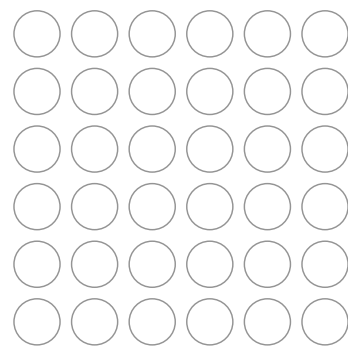
6



7



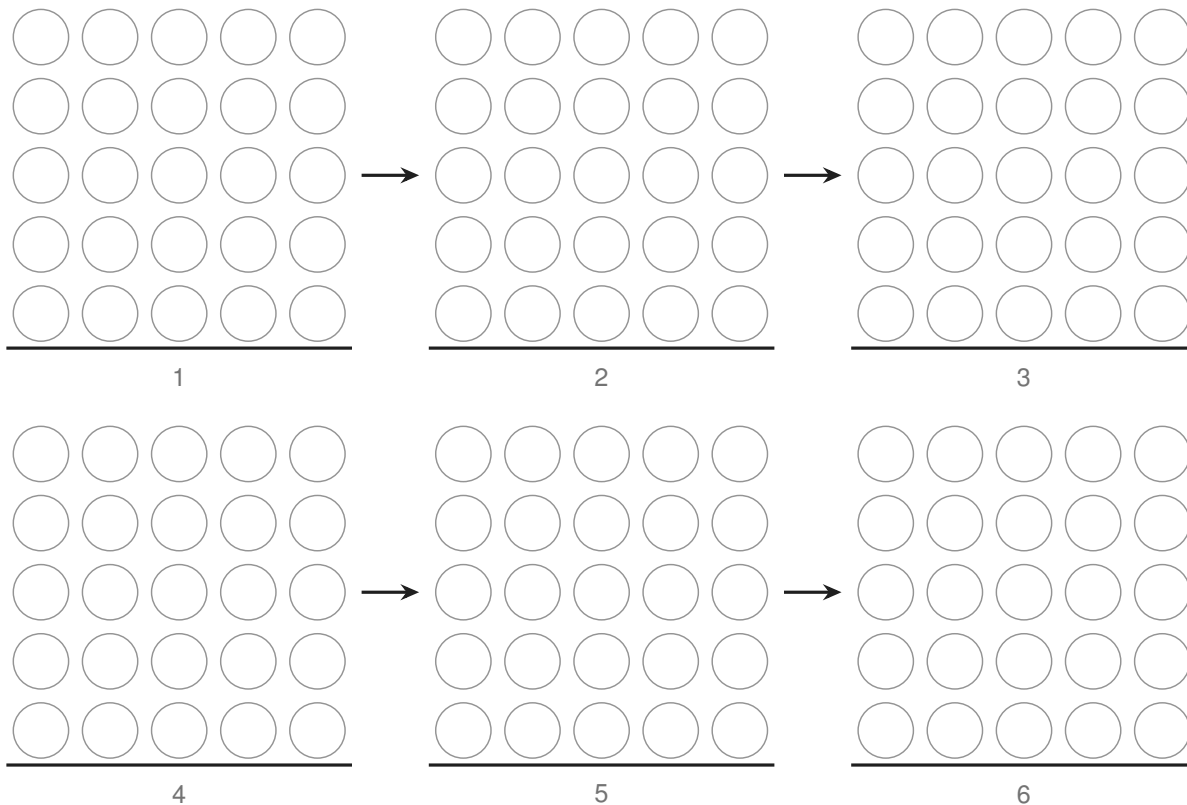
8



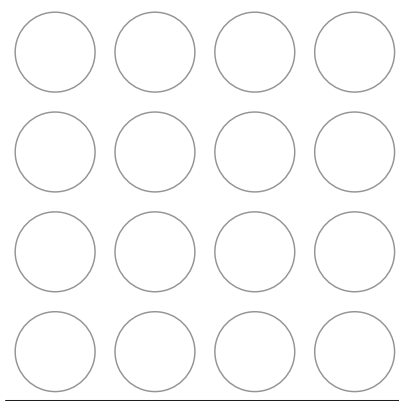
9



Problem 10: Take 5 counters. Start with all 5 in one pile and color it in picture 1. Do the move and color, until all six pictures are colored. Circle two pictures that look the same. If one of your pictures is a stairs shape, with piles 1, 2, 3 and so on, draw a star next to it.



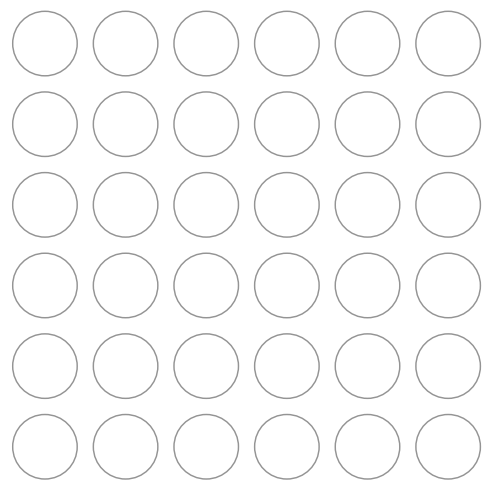
Problem 11: Build stairs with four steps: piles of 4, 3, 2 and 1. Count all the counters and write the number in the box. Do the move. Put the piles in a row, biggest first, and color them in the picture.



Problem 12: Perfect stairs means piles of 1, 2, 3 and so on, one pile of each size, with no counter left over. Can 7 counters make perfect stairs? Try every number from 1 to 12. Circle each number that makes perfect stairs. Cross out each number that does not.

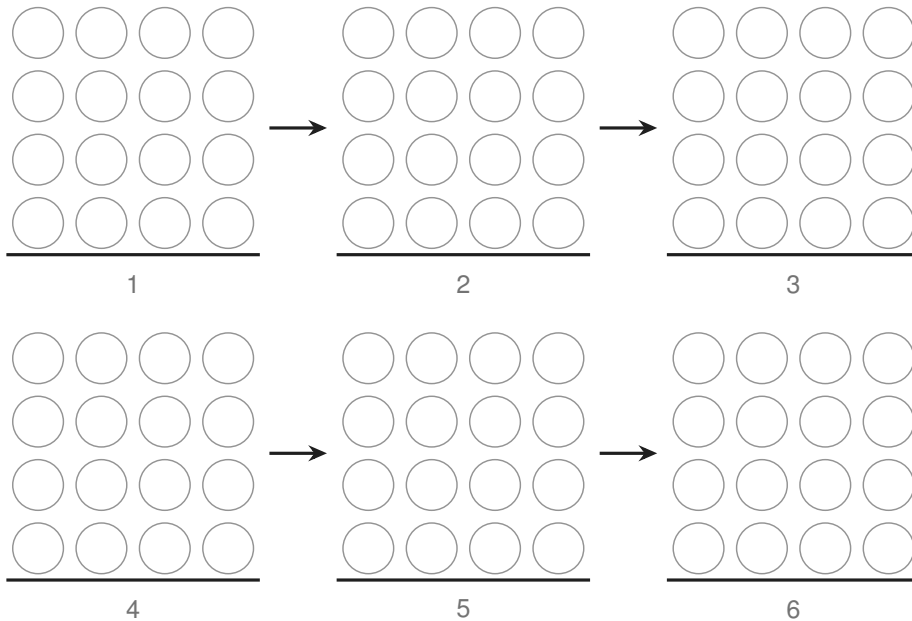
1	2	3	4	5	6
7	8	9	10	11	12

Problem 13: Build stairs with five steps and color them in the picture. Count the counters and write the number in the first box. Then build stairs with six steps, count the counters, and write the number in the second box.



--	--

Problem 14: Take 7 counters. Build the stairs 3, 2, 1 and put the extra counter on top of the biggest pile, so the piles are 4, 2, 1. Color this in picture 1. Do the move and color, until a picture looks exactly like one you colored before. Circle the two matching pictures. In every picture, use one color for the stairs 3, 2, 1 and a different color for the extra counter.



Problem 15: Take 11 counters. Build the stairs 4, 3, 2, 1 and put the extra counter on top of the biggest pile, so the piles are 5, 3, 2, 1. Color this in picture 1. Do the move and color, until a picture looks exactly like one you colored before. Circle the two matching pictures and count the arrows between them.

